

The shearing box, and the three numbers that govern it

Why the global runs cannot resolve the MRI; what Pm, Rm and α are; and what a scan over Pm measures

Report III — companion to `report_rot192_rot256` and `report_late_convergence`

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Abstract

Reports I and II found that the differential rotation of a $2.005 M_\odot$ magnetised white dwarf is not erased, and steepens by 22.6%. Both are bounded by one limitation: the magnetorotational instability is not resolved, with a quality factor $Q = \lambda_{\text{MRI}}/\Delta x \simeq 0.4$ over the very window in which the steepening is measured. This report sets out the route around that limitation and the first results along it. A *local* calculation can reach what the global grid cannot, because in the incompressible approximation — valid here to $v/c_s \sim 4 \times 10^{-4}$ — the acoustic timestep disappears and a hundred orbits costs hours rather than months. Three dimensionless numbers govern what such a box can say, and all three are defined and evaluated here for the first time in this campaign. **The magnetic Prandtl number of the stellar interior is $\text{Pm} \approx 750$** , computed from electron transport in degenerate matter — not the $\text{Pm} \sim 0.6$ of the cool crystallising white dwarfs in the literature, and two orders *above* rather than below the threshold at which MRI turbulence is known to die. **The magnetic Reynolds number is the campaign’s central limitation:** the star sits at $\text{Rm} \sim 10^{20}$ and our global runs at $\text{Rm} \approx 5$, a gap of twenty orders of magnitude that no resolution we can afford closes. A scan of five box simulations over $\text{Pm} = 1\text{--}16$ finds transport rising as $\text{Pm}^{0.19}$ and then flattening above $\text{Pm} = 8$; whether that flattening is physical or a resolution artefact is not settled here. Either way the transport at the stellar Pm is 1.6–3.7 times the $\text{Pm} = 1$ value — a factor of a few, not orders of magnitude, so it is α and not the Prandtl number that decides the outcome. Converting the measured stress to the star gives $\alpha \sim 8 \times 10^{-4}$. An envelope estimate put the braking time at 130 s against a 78 s run; replacing it with an actual torque across the half-mass radius gives 293–557 s, four times longer, so the rotation would *not* have been erased inside our run. What replaces that claim is sharper: **the predicted MRI torque is 1.3 times the angular-momentum transport we actually observe, and opposite in sign.** The observed transport is inward and produces the steepening; the MRI would drive it outward. It is a leading-order competing effect of the same size, and whether the steepening survives it cannot be settled by anything measured here.

1 Why a box at all

Report I states on its first page that the MRI is unresolved. The number is worth repeating because everything here follows from it. Measured from the volume-typical vertical field rather than its peak, the quality factor $Q = \lambda_{\text{MRI}}/\Delta x$ peaks at 6.5 near $t = 4.5$ s and sits at 0.3–0.5 over $t = 40\text{--}78$ s. Representing the linear MRI mode needs $Q \gtrsim 6$; converged turbulence needs $Q \gtrsim 15\text{--}20$, which this mesh never approaches under any assumption.

Refining is not the answer. Cost scales as N^4 — cells as N^3 and timesteps as N — so the factor needed is prohibitive. What changes the problem is dropping to a *local* model and, with it, changing the equations.

The incompressible approximation, and why it is not a compromise here. In incompressible MHD the thermal pressure becomes a Lagrange multiplier enforcing $\nabla \cdot \mathbf{v} = 0$, the sound speed is formally infinite, and the acoustic constraint $\Delta t \leq \text{cfl } \Delta x / c_s$ vanishes. The approximation is valid when both the flow and the Alfvén speed are far below c_s . In our interior $v/c_s \sim 4 \times 10^{-4}$ and $v_A/c_s \sim 10^{-5}$, so it is not marginally acceptable but ideal. The consequence is a factor $c_s/v_{\text{turb}} \sim 5 \times 10^4$ in the number of timesteps per orbit: a compressible box needs 5×10^6 steps per orbit, an incompressible one about 100. Measured throughput is 80 s per orbit at 64^3 on eight threads, so a hundred orbits is about two hours.

A second consequence matters for interpretation: **the plasma β does not appear in the incompressible equations at all.** The $\beta \sim 10^9$ that made a compressible box unaffordable was a property of the numerical method we had chosen, not of the physics. What remains is the shear parameter q , the box size in units of λ_{MRI} , and the numbers below.

What the box cannot do. The Tayler instability, which is what destroys the ordered field in the global runs, is driven by the curvature of toroidal field lines about the rotation axis. A local Cartesian box has no curvature, so the $m = 1$ kink cannot grow there by construction. Nothing in this report bears on the field-destruction result of Reports I and II.

2 The three numbers

2.1 Pm: which diffusion wins

The magnetic Prandtl number is the ratio of two diffusivities, both in $\text{cm}^2 \text{s}^{-1}$:

$$\text{Pm} = \frac{\nu}{\eta}, \quad (1)$$

with ν the kinematic viscosity — how fast *momentum* spreads — and η the magnetic diffusivity — how fast the *field* slips through the fluid. It says nothing about how viscous or how resistive the material is, only which of the two acts faster.

The physics is in which dissipation bites first as one goes to small scales. At $\text{Pm} \gg 1$ viscosity smooths the velocity field while the magnetic field survives, so the field reaches scales *smaller* than the motions that made it. At $\text{Pm} \ll 1$ the field leaks away before the velocity is damped, and one can have vigorous turbulence that sustains no field at all — which is why liquid-sodium laboratory dynamos are so difficult. For the MRI this is decisive, because the instability manufactures small-scale structure: whether it sustains itself, and how much angular momentum it carries, depends on which dissipation arrives first.

2.2 Rm: whether the field is carried or leaks

The magnetic Reynolds number compares advection of the field against its diffusion,

$$\text{Rm} = \frac{LV}{\eta}, \quad \text{alongside} \quad \text{Re} = \frac{LV}{\nu}, \quad \text{so that} \quad \text{Pm} = \frac{\text{Rm}}{\text{Re}}. \quad (2)$$

At $\text{Rm} \gg 1$ the field is frozen into the fluid in the sense of Alfvén’s theorem: field lines are dragged with the flow, and stretching or amplifying them is a matter of moving the fluid. At $\text{Rm} \lesssim 1$ the field diffuses out faster than the flow can move it, and no amount of stirring will hold it. Dynamos have a critical Rm , typically 10–100, below which no flow sustains a field.

2.3 α : how much angular momentum the turbulence actually carries

Pm and Rm say what regime the plasma is in. α is the answer one is after: a dimensionless measure of *transport efficiency*, introduced by Shakura & Sunyaev for accretion discs and used ever since as the single number summarising what turbulence does to angular momentum. It is the turbulent stress in units of the pressure,

$$\alpha = \frac{W_{\varpi\varphi}}{P} = \frac{\langle \rho \delta v_{\varpi} \delta v_{\varphi} \rangle - \langle B_{\varpi} B_{\varphi} \rangle / 4\pi}{P}, \quad (3)$$

Reynolds stress plus Maxwell stress over pressure, and it enters the dynamics as an effective viscosity $\nu_t = \alpha c_s H$. Resolved MRI simulations find $\alpha \sim 0.01$ without net flux and $\alpha \sim 0.1$ – 1 with it.

Two things about α matter here specifically.

An incompressible box cannot give α directly. There is no pressure in the incompressible equations, so the denominator does not exist. What the box measures is the stress normalised by the imposed field, W/B_0^2 , and converting to α requires the star’s own field and pressure:

$$\alpha = \frac{2}{\beta} \frac{W}{B_0^2}, \quad \beta = \frac{P}{B_z^2 / 8\pi}. \quad (4)$$

So α is small in our star not because the MRI is inefficient but because the vertical field is weak relative to the pressure.

α is what sets the answer, and Pm only modifies it. The braking time of the differential rotation follows from ν_t , so α is the quantity that decides whether the rotation survives. The Prandtl number enters only through the factor of a few by which it shifts α — which is why the modest Pm dependence measured in Section 5 is good news rather than a disappointment.

2.4 Why Rm is this campaign’s central limitation

Table 1 is, in one place, the reason the global results are bounded.

		Rm
The star	degenerate matter, $\eta = 3.3 \times 10^{-3} \text{ cm}^2/\text{s}$	$\sim 10^{20}$
Our 192 ³ run	set by the mesh	≈ 3
Our 256 ³ run	set by the mesh	≈ 6
The box, this report	set by us, explicitly	10^3 – 1.6×10^4

Table 1: The star is effectively perfectly conducting, with an ohmic decay time of order 10^{10} years. Our global simulations behave as though the matter were twenty orders of magnitude more resistive. This is not a modelling choice: the mesh itself diffuses field, with a numerical diffusivity $\eta_{\text{num}} \approx 0.6$ – $0.9 c_s \Delta x$ extracted from the measured decay, and it cannot be switched off.

Two things follow. First, at $\text{Rm} \approx 5$ the global runs are not in quasi-ideal MHD but in a *diffusive* regime, so “the ordered field is destroyed” mixes a real $m = 1$ instability with mesh leakage, and two resolutions cannot separate them. Second, this is why a large-eddy treatment was rejected: such models represent a turbulent cascade continuing below the cell, and at $\text{Rm} \approx 5$ there is no cascade to represent.

The box inverts the situation. There Rm is *explicit* — written into the input file and enforced by a physical resistivity well above the numerical one — which is precisely why a local calculation can measure what the global run cannot.

3 The stellar Pm

Electron transport dominates in degenerate matter. Following [Shternin \(2008\)](#), the shear viscosity is

$$\eta_{\text{visc}} = \frac{n_e v_F p_F}{5 \nu_e}, \quad \nu_e = \nu_{ee} + \nu_{ei}, \quad (5)$$

and the asymmetry that fixes Pm is that electron–electron collisions conserve the total electron momentum. They therefore do *not* degrade a current and stay out of the electrical conductivity, while they *do* degrade shear stress and enter the viscosity. Hence $\sigma = n_e e^2 / (m^* \nu_{ei})$ with ν_{ei} alone, and $\eta = c^2 / 4\pi\sigma$.

At the run’s mean density $\rho = 4.8 \times 10^7 \text{ g cm}^{-3}$ and an assumed remnant temperature $T = 10^8 \text{ K}$, this gives $\nu = 2.4$ and $\eta = 3.3 \times 10^{-3} \text{ cm}^2 \text{ s}^{-1}$, so

$$\boxed{\text{Pm} \approx 750.} \quad (6)$$

The result is robust: Pm stays above 190 across $\rho = 5 \times 10^7$ – 10^9 and $T = 10^7$ – 10^9 K , and above 30 even for a Coulomb logarithm of 5. Reaching Pm = 1 would need a Coulomb logarithm near 27, which is unphysical. The ions are a liquid throughout ($\Gamma = 13$, melting near 175), and electron–ion scattering dominates ($\nu_{ee}/\nu_{ei} = 0.04$).

This corrects a value taken from the literature. A figure of $\text{Pm} \sim 0.6$ circulates for white dwarfs and was used earlier in this campaign. It is quoted for the convective region of a *cool crystallising* CO white dwarf — a much lower density and temperature, and plausibly a regime where ion rather than electron viscosity dominates. It does not describe a hot $2 M_\odot$ merger remnant.

The correction matters twice over, and in our favour both times. The threshold below which MRI turbulence dies, $\text{Pm}_{\text{crit}} \sim 2$ –4, is no longer a concern: we are two to three orders *above* it. And we sit in the same high-Pm regime as the protoneutron-star boxes in which published subgrid closure coefficients were calibrated, so those coefficients have a far better chance of transferring than had been feared.

One assumption is unavoidable and should be visible. The equation of state used in the global runs is barotropic and carries no temperature at all, so T here is an assumption about a merger remnant rather than a simulation output. That is why the calculation is presented as a scan rather than a single value.

4 The scan

Five incompressible shearing-box simulations with SNOOPY v6.0 ([Lesur & Longaretti 2007](#)), 64^3 , box (4, 4, 1), net vertical flux $B_{z0} = 0.1$, $\text{Re} = 1000$ fixed and Pm varied from 1 to 16, each run to 31.8 orbits with 1401 samples averaged over $t > 60$.

Keplerian shear $q = 1.5$ was used deliberately, rather than our star’s $q = 0$ –2. It is where $\alpha(\text{Pm})$ has been published, so the same runs measure the trend and check our use of the code against numbers obtained independently. Scanning our own q comes after that check; the other order would leave any disagreement unattributable between a setup error and physics.

SNOOPY works in Alfvén units, so b is v_A and the total angular-momentum flux is $W = \langle v_x v_y \rangle - \langle b_x b_y \rangle$, Reynolds plus Maxwell, the Maxwell term entering with a minus sign. There being no sound speed in an incompressible box, the usual $\alpha = W/c_s^2$ is meaningless and the net-flux normalisation W/B_0^2 is used instead.

Pm	W/B_0^2	Maxwell	Reynolds	Max/Rey
1	19.32 ± 0.21	0.160	0.0335	4.76
2	22.67 ± 0.26	0.191	0.0362	5.26
4	25.24 ± 0.30	0.215	0.0374	5.74
8	31.84 ± 0.33	0.271	0.0479	5.65
16	31.88 ± 0.32	0.272	0.0473	5.74

Table 2: Transport against Pm. The last two rows are indistinguishable. Maxwell/Reynolds between 4.8 and 5.7 is above the 3–5 of zero-net-flux boxes and is what net flux is expected to give; it is a check on the setup, not a result.

4.1 The fit, and a correction to our own pre-registered criterion

The five points fit $W/B_0^2 = 19.7 \text{ Pm}^{0.194}$.

Before the runs started, a sanity criterion was registered: the literature finds slopes of 0.5–1, so a slope near zero or negative would indicate a setup error rather than interesting physics. **That criterion was wrong.** Lesur & Longaretti (2007) give δ in the range 0.25–0.5 over $0.12 < \text{Pm} < 8$ and $200 < \text{Re} < 6400$. The 0.5–1 had been taken from a search summary and recorded as though verified.

Against the correct band our 0.19 sits just below — same sign, same order. That is acceptable for a single unvalidated resolution, but it is not agreement and is not reported as such.

4.2 The flattening, which is the real finding

$\text{Pm} = 8$ and $\text{Pm} = 16$ agree to 0.039 against a 2σ of 0.92. Two readings are consistent with the data and it does not separate them:

1. $\alpha(\text{Pm})$ genuinely saturates above $\text{Pm} \sim 8$. This is plausible and untested — Lesur & Longaretti (2007) stopped at exactly $\text{Pm} = 8$.
2. At 64^3 numerical resistivity already dominates, so the code cannot distinguish $\text{Rm} = 8000$ from $\text{Rm} = 16000$. This possibility was recorded before any numbers existed, and it fired.

The test that separates them is one run at 128^3 and $\text{Pm} = 16$: if W rises, the 64^3 point was resolution-limited. It costs roughly $16\times$ a 64^3 run and has not yet been done.

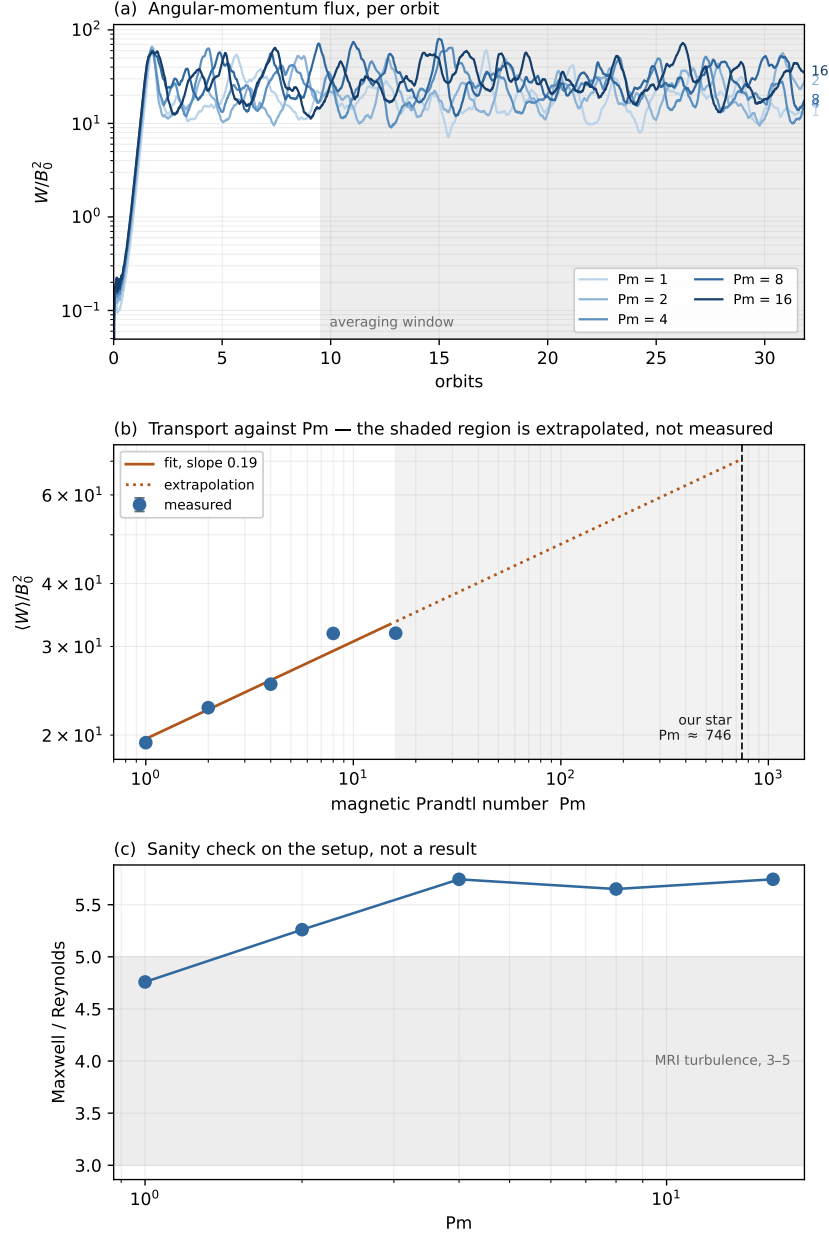


Figure 1: (a) All five saturate within about two orbits and sustain turbulence for thirty; the shaded band is the averaging window. (b) Transport rises as $Pm^{0.19}$, but the top two points coincide; the shaded region to the right is *extrapolated, not measured*. (c) The Maxwell/Reynolds ratio, a check on the setup rather than a result. Data: `shearing_box/analyse_pm_scan.py`.

5 The implied braking time, and why it is uncomfortable

Equation (4) can be evaluated with the star’s own numbers, and the result is the sharpest statement this report contains. Using the measured late-time volume-typical poloidal field $B_{\text{pol,rms}} = 7 \times 10^{10} \text{ G}$, so $B_z \simeq 5 \times 10^{10} \text{ G}$, and the degenerate pressure at the run’s mean density, $P = 7.8 \times 10^{24} \text{ erg cm}^{-3}$:

5.1 Done properly: an actual torque

$H \sim R_{\text{eq}}$ was the crudest input in that chain and the conclusion is sharp enough to be worth replacing. The turbulent stress exerts a torque across the cylinder separating the inner half of the

plasma β on the vertical field	8×10^4
W/B_0^2 at the top of the scan	32
implied α	8×10^{-4}
turbulent viscosity $\nu_t = \alpha c_s H$, $H \sim R_{\text{eq}}$	$7.6 \times 10^{14} \text{ cm}^2/\text{s}$
braking time R_{eq}^2/ν_t	$\sim \mathbf{130 \text{ s}}$
length of our longest run	78 s

Table 3: Order of magnitude only — see the caveats below. The two rows that matter are the last two.

mass from the outer half, so the inner region redistributes its angular momentum on

$$\tau = \frac{L_{z,\text{inner}}}{G(r_{1/2})}, \quad G = 2\pi r_{1/2}^2 Z(r_{1/2}) W, \quad W = \left(\frac{W}{B_0^2}\right)_{\text{box}} \frac{B_z^2}{4\pi}, \quad (7)$$

with $L_{z,\text{inner}}$, $r_{1/2}$ and the shell definitions taken from the diagnostic itself and Z the star’s vertical extent there. A second improvement comes with it: $r_{1/2} = 0.26 R_{\text{eq}}$, where the Komatsu law gives $q = 0.47$, and the q scan measures $W/B_0^2 = 18.1$ there rather than the 32 of Keplerian shear.

This gives $\tau = 557 \text{ s}$, or 293–428 s once the Pm correction is folded in — **four times longer than the envelope**. So the sharpest version of the claim, that the rotation should already have been erased inside our run, does not survive doing the calculation properly.

5.2 What replaces it, and it is sharper

The comparison that decides the question is not against the star’s total angular-momentum loss, which is L_z leaving the star altogether and a different thing from internal redistribution. It is against the observed rate of change of the *inner* region’s angular momentum, since that and the MRI torque are both the rate at which L_z crosses $r_{1/2}$.

over $t = 30\text{--}78 \text{ s}$		
observed $dL_{z,\text{inner}}/dt$	$+4.21 \times 10^{46}$	erg
observed $dL_{z,\text{outer}}/dt$	-1.35×10^{47}	erg
predicted MRI torque G	5.59×10^{46}	erg

Table 4: The inner region *gains* angular momentum while the outer loses it: transport is inward, which is the steepening. The MRI would drive it outward.

The predicted MRI torque is 1.3 times the observed inward transport, and opposite in sign. That is the result: not that the MRI would have erased the differential rotation, but that it is a *leading-order competing effect* of the same size as the transport producing the steepening, pointing the other way. Switching it on could plausibly cancel the steepening rather than merely dent it, and the sign of the outcome is not predictable from anything measured here.

This is consistent with [Miravet-Tenés et al. \(2025\)](#), who switch on a subgrid MRI closure in a differentially rotating magnetised neutron star and find the inner profile flattening where we find it steepening.

Two things the late window adds. Past 60 s the inner region stops gaining — $dL_{z,\text{inner}}/dt$ falls to -1.5×10^{44} , essentially zero — while the outer keeps losing. The steepening continues at half its earlier rate, -0.32 against $-0.62\% \text{ s}^{-1}$, but it is now driven by the outer region shedding angular momentum rather than the inner gaining it. Those are different mechanisms and only the first competes directly with the MRI.

What still limits the estimate. B_z is taken uniform at the volume-typical value, and $\alpha \propto B_z^2$; if the field is weaker at $r_{1/2}$ than on average, the torque falls and τ rises. $r_{1/2}$ is imported from the 192³ diagnostic, which carries it where the 256³ CSV does not. And the field itself is the decayed late-time one, which Report II shows is not converged. A factor of a few either way remains.

6 What this says about the star, and what it does not

Relative to $\text{Pm} = 1$, the transport at the stellar $\text{Pm} \approx 750$ is a factor 3.7 if the trend continues and 1.6 if it has already saturated. **A factor of a few in either case, not orders of magnitude.** That is itself useful: what decides the fate of the differential rotation is the value of α , not the Prandtl number, and the high-Pm correction is a modest one.

Three limitations bound all of the above.

1. **The extrapolation is real and is 1.7 decades wide.** No direct numerical simulation reaches $\text{Pm} = 750$, which would need $\text{Rm} = 750 \text{Re}$ and so $\text{Rm} \sim 10^6$ at any resolvable Re . The box brackets rather than matches. It is a lower bound only if the trend does not saturate — and our own top two points suggest it may.
2. **The box does not remove extrapolation, it moves it** from β to (Rm, Pm) . A direct simulation reaches $\text{Rm} \sim 10^3$ – 10^4 against the star’s 10^{20} . The gain is that this is one extrapolation in a named parameter whose direction is known, instead of several that were not.
3. **Nothing here touches the field.** The box has no curvature and so no Tayler instability, and the subgrid model these coefficients would feed enters the momentum equation only, not the induction equation. The destruction of the ordered field reported in Reports I and II remains without a counterpart calculation.

Next

1. 128³ at $\text{Pm} = 16$, to decide whether the flattening is physics or resolution. It is the one result above that is currently ambiguous.
2. Scan q over 0–2, the range our Komatsu rotation law spans, now that the Keplerian check has been made.
3. Scan the net toroidal-to-vertical flux ratio, which in our star runs from 2 to 9.5 in amplitude. This is the configuration MInIT’s vertical-field MRI does not obviously describe, and the box can test which instability dominates at our ratio.
4. Extract the closure coefficients and compare with the published $\alpha^{\text{PI}} = -1.4$, $\beta^{\text{PI}} = -0.8$.

Code, inputs and analysis are under version control in the `WD_MAG` repository, under `shearing_box/`. SNOOPY v6.0 is GPL-3.0 and is not redistributed there; `build.sh` reproduces the build, including its FFTW dependency, without root.

References

Lesur, G., & Longaretti, P.-Y. 2007, *Impact of dimensionless numbers on the efficiency of magnetorotational instability induced turbulent transport*, MNRAS, 378, 1471.

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